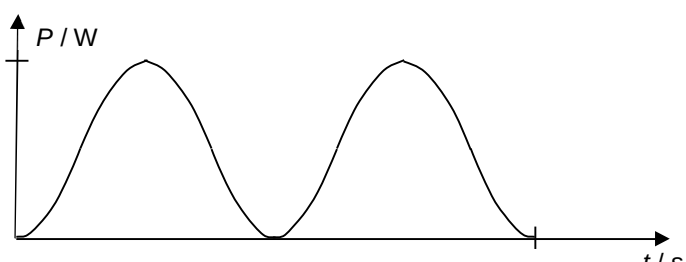


	$\lambda = 3.75 \times 10^{-11} \text{ m}$	
7(a)	The root-mean-square (r.m.s.) value of an alternating current is defined as the value of <u>steady direct current</u> which would <u>produce the same mean power</u> as the alternating current <u>in a given resistance</u> .	[2] or zero
(b)	$\frac{N_s}{N_p} = \frac{V_s}{V_p}$ $\frac{N_s}{3000} = \frac{14}{250\sqrt{2}}$ $N_s = 119 \text{ turns}$ Assumption: the transformer is <u>ideal</u>, with no power loss in the stepping down process.	[1] for correct substitution [1] for correct answer [1]
(c)(i)	$\langle P \rangle = \frac{V_{rms}^2}{R} \leq 340$ $R \geq \frac{\left(\frac{14}{\sqrt{2}}\right)^2}{340} = 0.29 \, \Omega$	[1] for correct substitution [1] for correct answer
(ii)		[1] correct shape [1] correct number of cycles

9(a)(i)	Read off from Fig 9.1	
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